



Chapter 2 — Product Requirements & Feature Specifications

This chapter specifies all **10 core modules** of DOIP in functional detail: what each does, its user stories, functional requirements (FR), and acceptance criteria. Priorities use MoSCoW: **M**ust, **S**hould, **C**ould.

2.1 Module Overview & Priorities

#	Module	Priority	Phase
1	Command Dashboard	Must	1
2	Live Map	Must	1
3	Simulation Engine (see Ch. 7)	Must	1
4	Drone Mission Planner	Must	2
5	AI Situation Report	Must	3
6	RAG Knowledge Assistant	Must	3
7	Threat Detection (Computer Vision)	Should	4
8	Logistics Dashboard	Should	5
9	Personnel Health Monitor	Could	5
10	Weather Intelligence	Should	6
11	Predictive Analytics	Should	6

2.2 Command Dashboard

Purpose: The operator's home screen — a single-glance overview of the entire simulated operation.

User stories

- As an Operator, I want all critical information on one screen so I never have to context-switch during a run.
- As an Operator, I want alerts sorted by severity so I handle the most important item first.
- As an Exercise Director, I want to start, pause, and stop simulations from the dashboard.

Functional requirements

- FR-1.1: Display a grid of widgets: live map (center), active incidents, event feed, weather, resource levels, health alerts, mission timeline, AI recommendations.
- FR-1.2: All widgets update in real time via WebSocket (no manual refresh, no polling in the UI).
- FR-1.3: Widgets are collapsible and rearrangeable; layout persists per user.
- FR-1.4: Global search bar finds units, incidents, missions, and reports by name/ID.
- FR-1.5: Simulation controls: start / pause / resume / stop / speed multiplier (1x, 5x, 20x).
- FR-1.6: Severity color coding across the app: Critical (red), Warning (amber), Info (blue), Resolved (green).

Acceptance criteria: With a scenario running, every widget visibly updates within 1 second of the simulator emitting a relevant event; a paused simulation freezes all widgets consistently.

2.3 Live Map

Purpose: Geospatial heart of the platform, built on **MapLibre GL JS** + **OpenStreetMap** tiles.

User stories

- As an Operator, I want to see all units and incidents on a map that updates live.
- As an Operator, I want to toggle layers so I can declutter the view.
- As a Planner, I want to replay unit movement to review what happened.

Functional requirements

- FR-2.1: Pan/zoom map with OSM base tiles and optional terrain style.
- FR-2.2: Entity layers, individually toggleable: friendly units, drones (simulated), supply depots, medical teams, incidents, detections (CV), restricted zones, planned routes.
- FR-2.3: Smooth marker interpolation between position updates (no teleporting icons).
- FR-2.4: Click any entity → side panel with full details, recent events, and quick actions.
- FR-2.5: Heatmap layer for incident density; choropleth-style risk overlay per sector.
- FR-2.6: Time-slider **replay mode**: scrub back through the run and watch movement history.
- FR-2.7: Draw tools: polygons (zones), polylines (routes), circles (radii) — saved as GeoJSON.
- FR-2.8: Route visualization with per-segment risk coloring.

Acceptance criteria: 100+ moving entities render at 60 fps on a mid-range laptop; layer toggles react instantly; replay reproduces recorded movement faithfully.

2.4 AI Situation Report

Purpose: Convert a stream of raw simulated reports into concise, prioritized situation summaries.

Example transformation

Input events:

- "Drone Alpha observed vehicle movement."
- "Team Bravo delayed due to heavy rain."
- "Medical team requests evacuation."

AI output:

■ **Situation Summary — 14:32**

- Vehicle activity detected in Sector B (source: Drone Alpha).
- Weather is reducing movement speed; Team Bravo ETA +40 min.
- **Priority:** Medical evacuation request is unresolved — recommend tasking nearest medical unit.

Functional requirements

- FR-3.1: Generate a situation report on demand and automatically every configurable interval (default 5 sim-minutes).
- FR-3.2: Reports cite their source events (clickable links back to the event feed).
- FR-3.3: Reports are structured: Summary → Key developments → Risks → Recommended actions.
- FR-3.4: Operator can ask follow-up questions in a chat panel with full operational context.
- FR-3.5: All reports are stored and browsable as a timeline of the run.
- FR-3.6: A clear “AI-generated — verify before acting” label appears on every output.

Acceptance criteria: Given 20 mixed events, the report correctly prioritizes the critical one, references only real events (no hallucinated facts), and generates in under 15 seconds.

2.5 Sensor Fusion (Simulated)

Purpose: Combine multiple synthetic input streams into one coherent operational picture.

Simulated streams: GPS positions, camera detections (from CV service), environmental sensors, weather, and free-text unit reports.

Functional requirements

- FR-4.1: Every stream normalizes into the common Event schema (Chapter 6) with source, timestamp, geometry, and confidence.
- FR-4.2: Correlation rules group related events (e.g., a CV detection + a unit report in the same sector within 5 minutes → one “corroborated observation”).

- FR-4.3: Confidence scoring: corroborated events rank higher in feeds and AI summaries.
 - FR-4.4: The fused picture is queryable: "show all high-confidence events in Sector C in the last 30 minutes."
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2.6 Threat Detection (Computer Vision)

Purpose: Run object detection on **prerecorded, public aerial/disaster footage** and surface findings as map events. No live cameras.

Pipeline: Video file → frame sampler → YOLO inference → detected objects → risk scoring → geotagged events → map markers + event feed.

Detectable classes: vehicles, people, smoke, fire, damaged buildings/infrastructure.

Functional requirements

- FR-5.1: Upload or select a video; processing runs asynchronously with progress reporting.
- FR-5.2: Each detection stores class, confidence, bounding box, frame timestamp, and an assigned scenario location.
- FR-5.3: Risk score per detection = $f(\text{class weight, confidence, cluster density, proximity to units})$.
- FR-5.4: Detections above a risk threshold auto-create incidents.
- FR-5.5: Reviewer UI: filmstrip of annotated frames; operator confirms or dismisses detections.

Acceptance criteria: A 2-minute sample clip processes end-to-end and produces map-visible, reviewable detections with sensible risk ordering.

2.7 Drone Mission Planner (Simulated)

Purpose: Plan simulated aerial survey missions — zones, waypoints, timing, and battery budget — entirely in software.

Functional requirements

- FR-6.1: Draw a search zone polygon; the planner generates a lawnmower/spiral coverage path.
- FR-6.2: Manual waypoint editing: add, drag, reorder, delete.

- FR-6.3: Route optimization via **AI** (*grid/graph over the operational area*) with *Dijkstra** as baseline comparison; costs incorporate distance, restricted zones (hard constraint), and weather/risk penalties (soft constraints).
- FR-6.4: Mission estimate panel: total distance, duration at configured speed, simulated battery consumption with reserve margin, feasibility verdict.
- FR-6.5: Infeasible plans explain *why* (e.g., "exceeds battery by 18%; nearest recharge depot adds 2.4 km").
- FR-6.6: Launching a mission hands it to the simulation engine, which flies the virtual drone along the route in real time.

Acceptance criteria: For a given zone with one restricted area, the optimized route demonstrably avoids the restricted polygon and costs less than the naive route; battery model prevents impossible missions.

2.8 Logistics Dashboard

Purpose: Track and forecast consumables and equipment across the simulated force.

Tracked resources: fuel, food/water, medical kits, vehicle status, battery levels (simulated drones/equipment).

Functional requirements

- FR-7.1: Per-depot and per-unit inventory levels with history charts.
 - FR-7.2: Consumption is driven by simulation activity (movement burns fuel; incidents consume medical kits).
 - FR-7.3: Threshold alerts (warning at 30%, critical at 10%, configurable).
 - FR-7.4: **Shortage prediction:** projected time-to-empty per resource using current burn rate + analytics forecasts (Chapter 8).
 - FR-7.5: Resupply planning: create a resupply mission from depot to unit, integrated with the mission planner.
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2.9 Personnel Health Monitor

Purpose: Display synthetic wearable-style vitals for simulated personnel and raise alerts.

Synthetic signals: heart rate, body temperature, SpO₂, stress index — generated per person with realistic baselines, circadian drift, activity response, and injectable anomaly events.

Functional requirements

- FR-8.1: Roster view with live vitals and sparklines per person.
- FR-8.2: Threshold + trend alerts (e.g., "HR > 150 sustained 5 min", "SpO₂ falling").
- FR-8.3: Health alerts appear in the global event feed and on the map at the person's unit location.
- FR-8.4: Scenario hooks: an incident can inject health anomalies (e.g., heat exhaustion during a fire scenario).

2.10 Weather Intelligence

Purpose: Simulated weather system that visibly affects operations.

Functional requirements

- FR-9.1: Scenario-scripted or procedurally generated weather cells (rain, wind, fog/visibility, temperature) as moving polygons with intensity.
- FR-9.2: Map overlays: precipitation cells, wind vectors, visibility shading.
- FR-9.3: **Operational coupling:** weather modifies unit speed, route costs, mission feasibility, and sensor confidence — with the multipliers documented and visible in the UI.
- FR-9.4: Forecast panel: next 6 simulated hours per sector.

2.11 Predictive Analytics

Purpose: Forecast the near future of the operation.

Forecast targets & candidate models

Target	Model candidates	Output
Resource consumption	Prophet, XGBoost	Time-to-shortage per resource
Mission completion time	XGBoost regression	ETA + confidence interval
Vehicle maintenance risk	Gradient boosting classifier	Failure probability ranking
Supply needs next 24h	Prophet / LSTM	Reorder quantities

Functional requirements

- FR-10.1: Models train on accumulated simulation telemetry (every run enriches the dataset).
 - FR-10.2: Forecasts display with uncertainty bands — never as bare point estimates.
 - FR-10.3: Analytics page: per-run after-action report with charts (response times, resource efficiency, forecast accuracy).
 - FR-10.4: Model registry: version, training data window, and evaluation metrics per model.
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2.12 Cross-Cutting Requirements

Non-functional requirements

- NFR-1 **Latency**: simulator event → browser render < 1 s (p95).
- NFR-2 **Scale (demo)**: 200 concurrent simulated entities, 50 events/sec sustained.
- NFR-3 **Availability**: single-node deployment; clean restart recovers full state from DB.
- NFR-4 **Auditability**: every operator action and AI output is logged and replayable.
- NFR-5 **Accessibility**: keyboard navigation and colorblind-safe severity palette.
- NFR-6 **Internationalization-ready**: all UI strings externalized.

Out of scope (restated): hardware links, real camera/GPS feeds, weapons/targeting features, real-world operational data.